

IEOR 4101: Probability, Statistics and Simulation

Introduction and Overview

Henry Lam

Acknowledgements: Yuan Zhong and Karthyek Murthy

Motivation

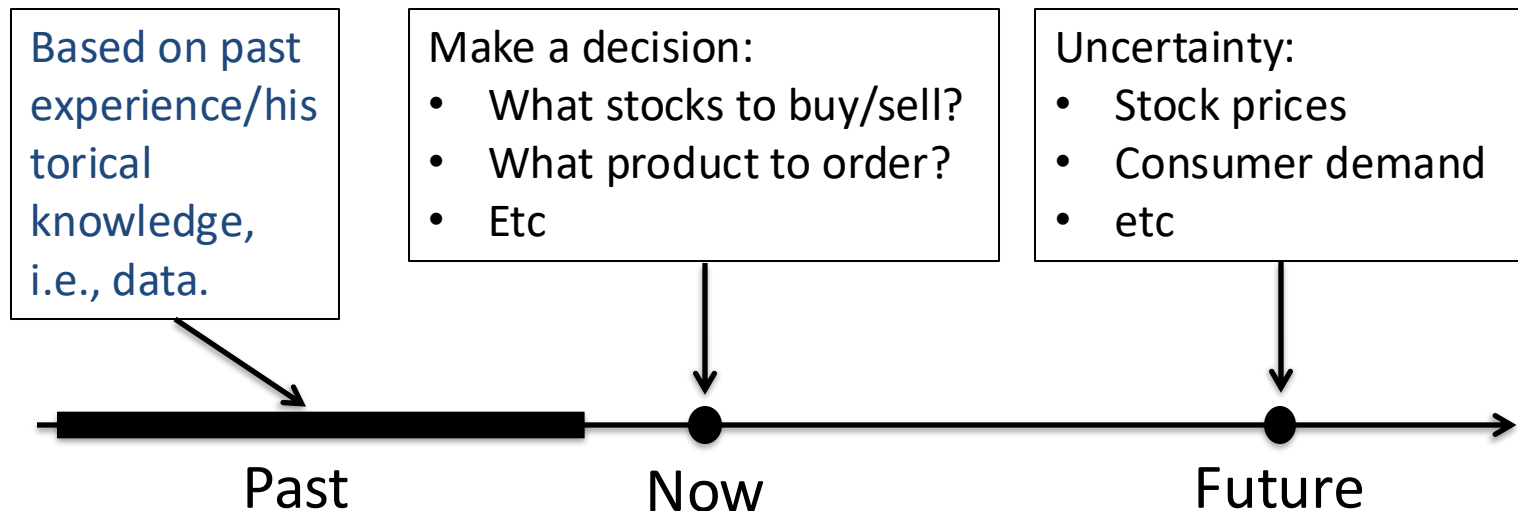
- Uncertainty is everywhere.
 - “There is a 30% chance that it will rain tomorrow.”
 - Who will win the US Open this year?
 - Stock price of Starbucks



– Etc.

Motivation

- As future business managers, you often need to ***make decisions under uncertainty***.
- Typical scenario:



Motivation

- This course is about making sense of data and uncertainty, and using this knowledge to aid decision making.
- Probability – To understand uncertainty (both from the past and for the future)
- Statistics – To extract information from data
- Simulation – A computation tool to assist decision making (with models, and also others)

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- You are an airline manager
- Managing a flight with 100 seats – revenue of \$200 each
- May be no-shows

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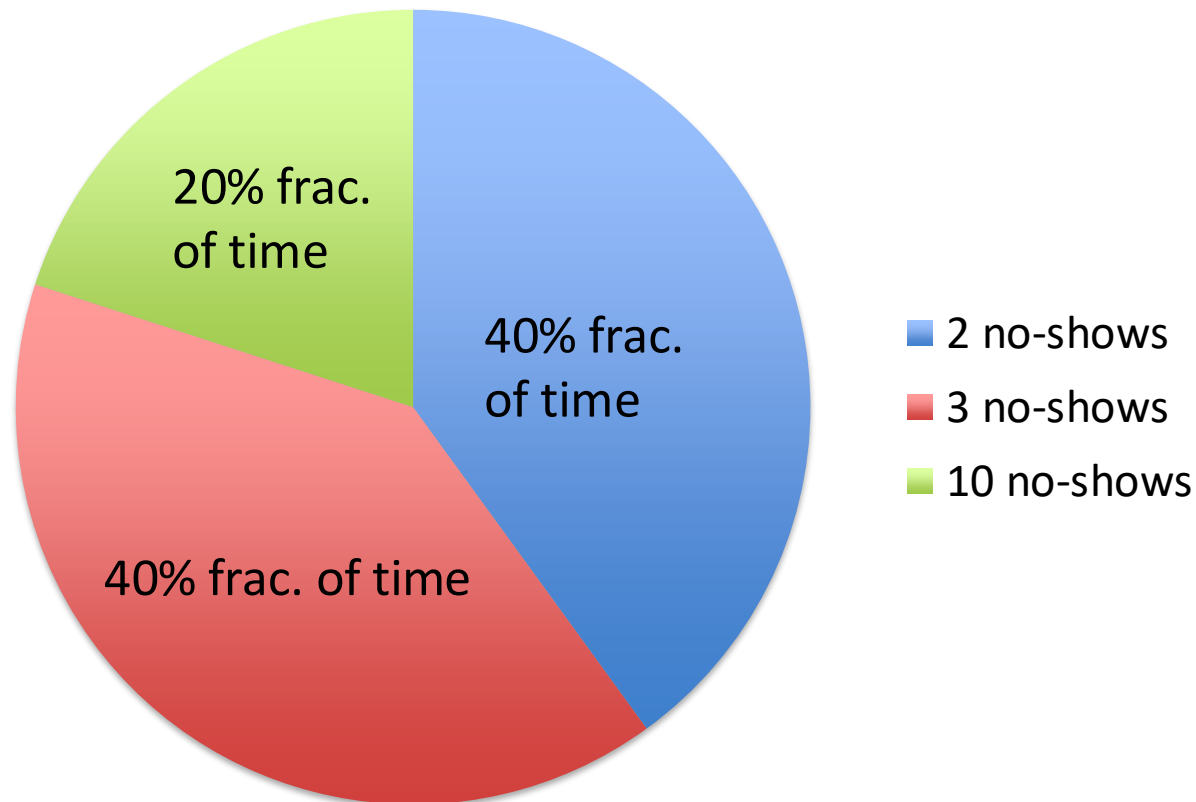
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1. Figure out the number of no-shows D .
 - Rarely a fixed number – a random quantity..
 - Understand this randomness based on **historical data**.

Motivation

Suppose the following distribution is available, obtained via:

- Statistical tools applied to the data directly, or
- A model using expert knowledge and/or data

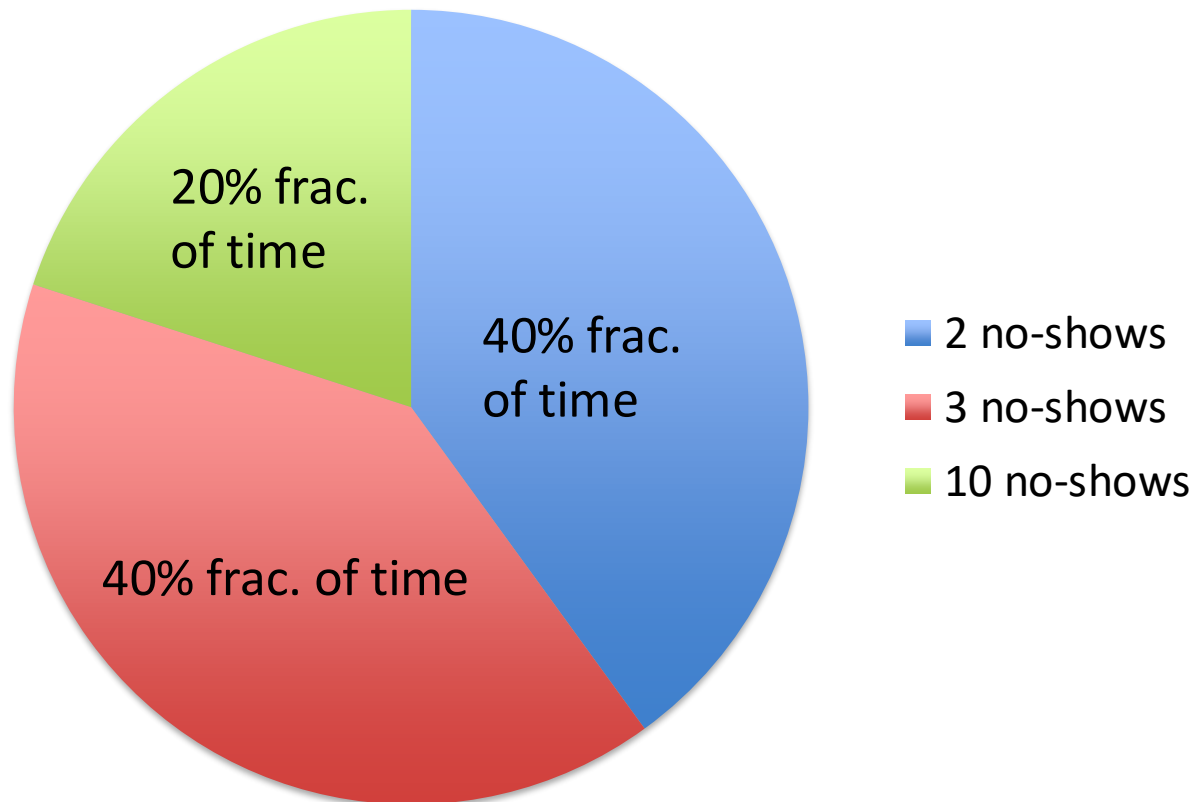


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1. Figure out the number of no-shows D .
 2. Next step?

Motivation

- “Average” # of no-shows = $2 \times 0.4 + 3 \times 0.4 + 10 \times 0.2 = 4$.
- Intuitively, oversell 4 seats?



Motivation

- “Average” # of no-shows = $2 \times 0.4 + 3 \times 0.4 + 10 \times 0.2 = 4$.
- If I oversell 4 seats (an intuitive solution):
 - If 2 no-shows, then revenue = $\$200 \times 104 - \$300 \times 2 = \$20,200$
 - If 3 no-shows, revenue = $\$200 \times 104 - \$300 = \$20,500$
 - If 10 no-shows, then revenue = $\$200 \times 104 = \$20,800$
- Total *average* revenue = $\$20,200 \times 0.4 + \$20,500 \times 0.4 + \$20,800 \times 0.2 = \$20,440$.

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- Total *average* revenue = $\$20,200 \times 0.4 + \$20,500 \times 0.4 + \$20,800 \times 0.2 = \20440 .
- Oversell 3 seats? *Average* revenue = $\$20480 > \20440 .
- Oversell 2 seats? *Average* revenue = $\$20400$

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- **Oversell 3 seats?** *Average* revenue = **$\$20480$** > \$20440.
- Oversell 2 seats? *Average* revenue = \$20400
- Optimal is to oversell 3 seats

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1. Figure out the number of no-shows D .
 2. Analyze the decision-making problem.
 3. Often times the problem is much more complicated than this example – resort to other means such as **simulation**

Motivation

- Probability – To understand uncertainty of demand
- Statistics – To extract information from historical demand data
- Analytical tool or simulation – To assist decision making

Motivation

- To summarize:

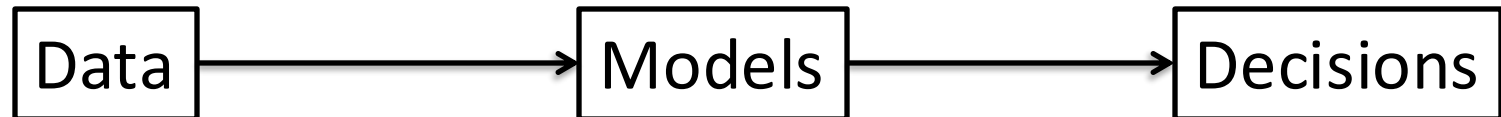


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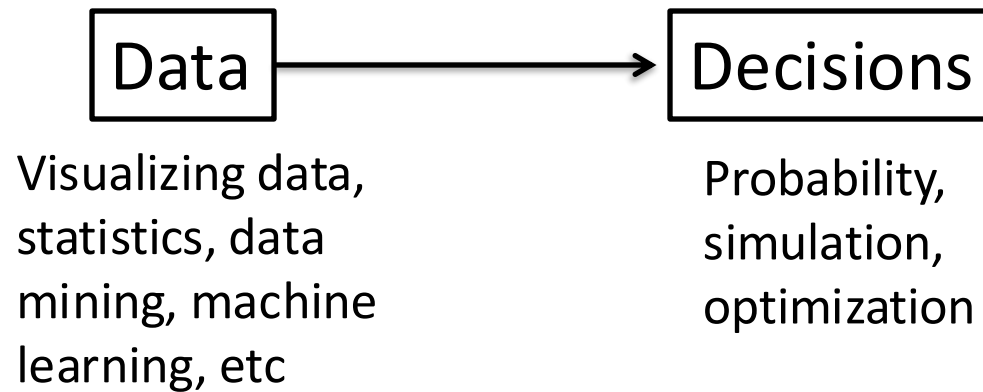


- Sometimes

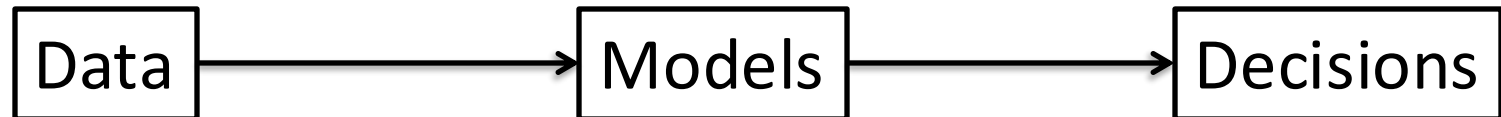


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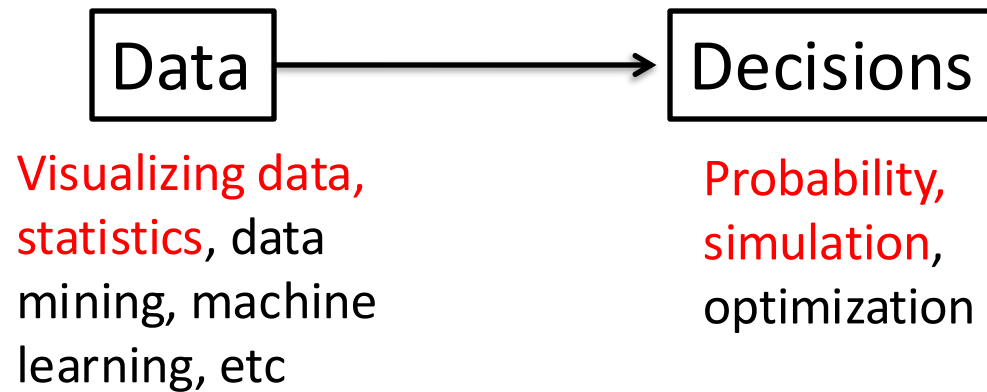


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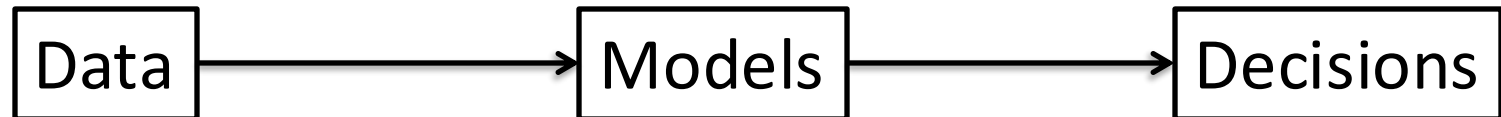


Motivation

- To summarize:



- Sometimes



Topics

- Probability
 - Probability space, conditional probability and independence, decision trees, Bayes' formula
 - Random variables (discrete and continuous), expectation, variance, covariance, correlation, common distributions
 - Law of large numbers, central limit theorem
- Simulation
 - Basic methods to simulate random variables
- Statistics:
 - Parameter estimation – interval and point estimation, confidence interval, maximum likelihood
 - Hypothesis testing; testing concerning mean/variance of population(s)
 - Regression (if time permits)

Topics

- Descriptive statistics (briefly)
 - Visualizing data, summarizing data
 - Histograms, pie charts, bar charts
 - Sample mean, median, mode, variance

What Are Data?

- Data can be just about any type of information
- Data can be quantitative/numerical:
 - Weights, heights, salaries, transactions, etc
- Data can also be qualitative/non-numerical:
 - Colors, locations, genders, etc

What Are Data?

- Depending on how data are used..
 - Example: “studies have shown that smoking increases the risk of cancer”
 - This is an example of *inferential statistics*
 - To draw sensible conclusions from data, we first need to describe them well – *descriptive statistics*
- We will first see some effective ways to *summarize* and *display* data

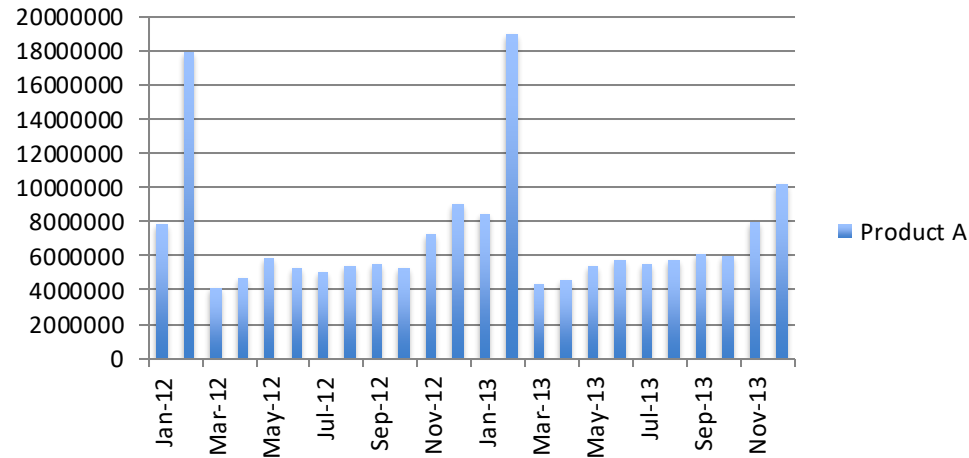
Bar Charts and Time Series Plots

- Consider the data on the right
- Sales data (in dollars) of two different chocolate candy products A and B
- One is Snickers Bar and One is Hershey's Kisses
- Q: Can you tell which is which?

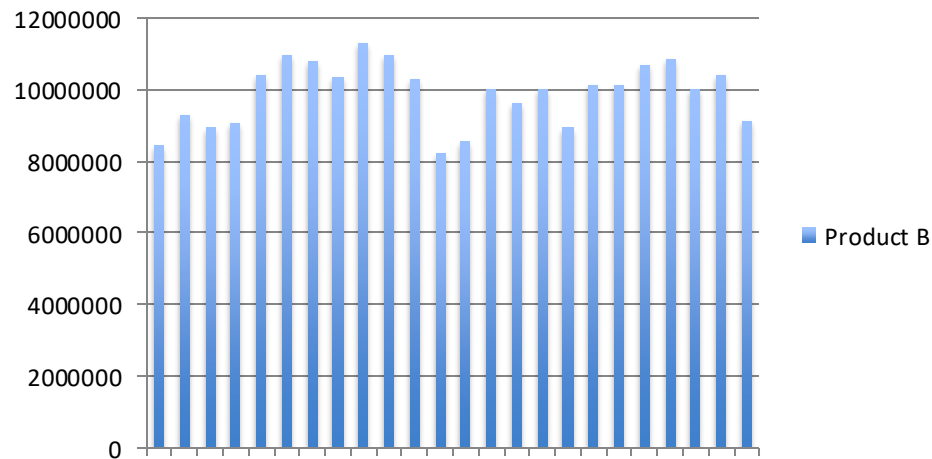
	Product A	Product B
Jan-12	7816658	8465858
Feb-12	17865227	9307039
Mar-12	4128771	8945254
Apr-12	4672123	9056663
May-12	5877760	10390335
Jun-12	5324896	10944793
Jul-12	4998175	10794320
Aug-12	5354155	10361255
Sep-12	5502833	11306859
Oct-12	5318757	10972435
Nov-12	7212337	10294797
Dec-12	9074508	8227434
Jan-13	8401754	8586088
Feb-13	18990757	10026389
Mar-13	4391132	9616869
Apr-13	4617467	10024582
May-13	5440681	8927481
Jun-13	5753250	10115303
Jul-13	5523823	10135208
Aug-13	5782933	10674501
Sep-13	6113621	10831740
Oct-13	6016061	10013292
Nov-13	7983319	10380192
Dec-13	10192890	9133822

Bar Charts and Time Series Plots

Product A

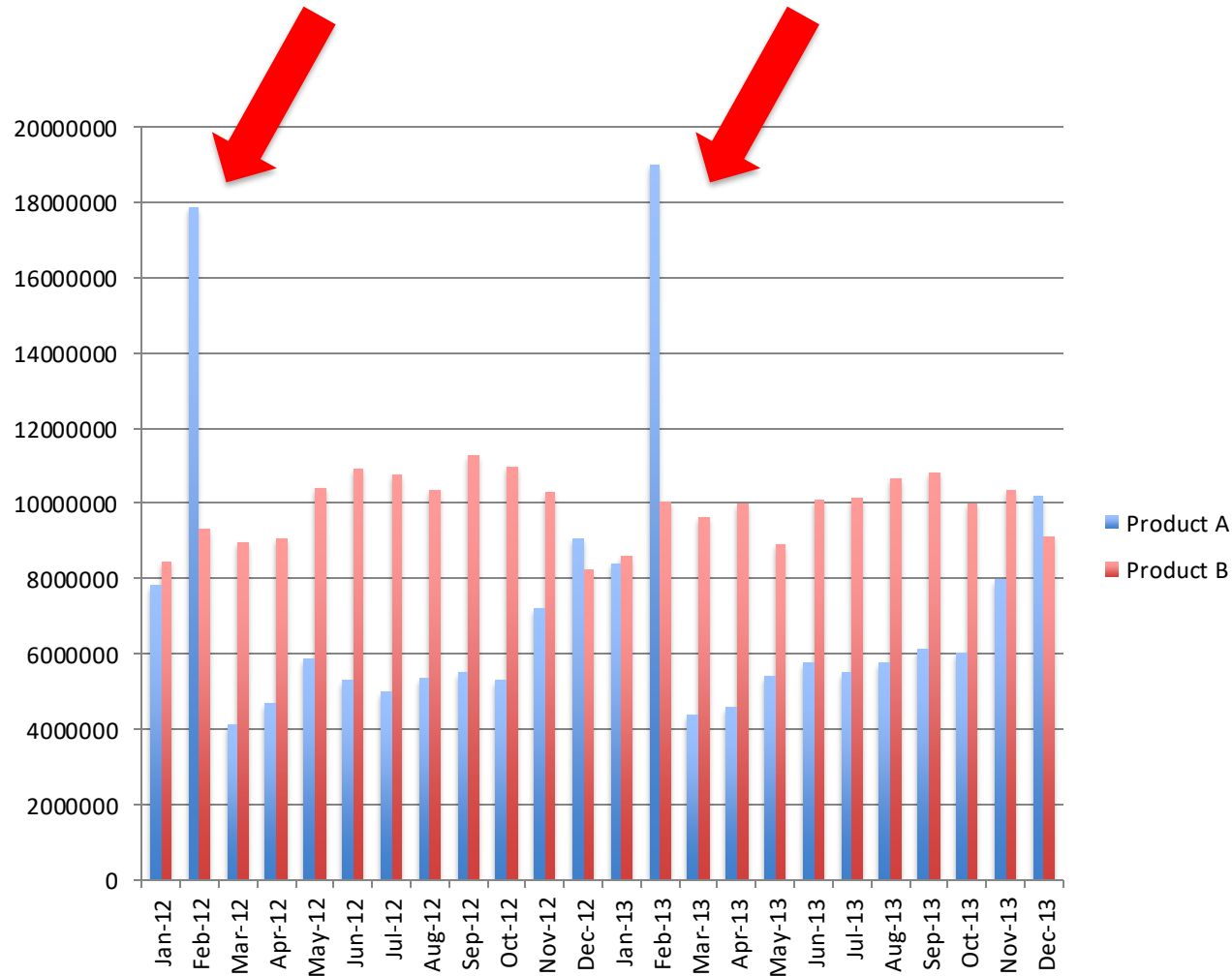


Product B



Bar Charts and Time Series Plots

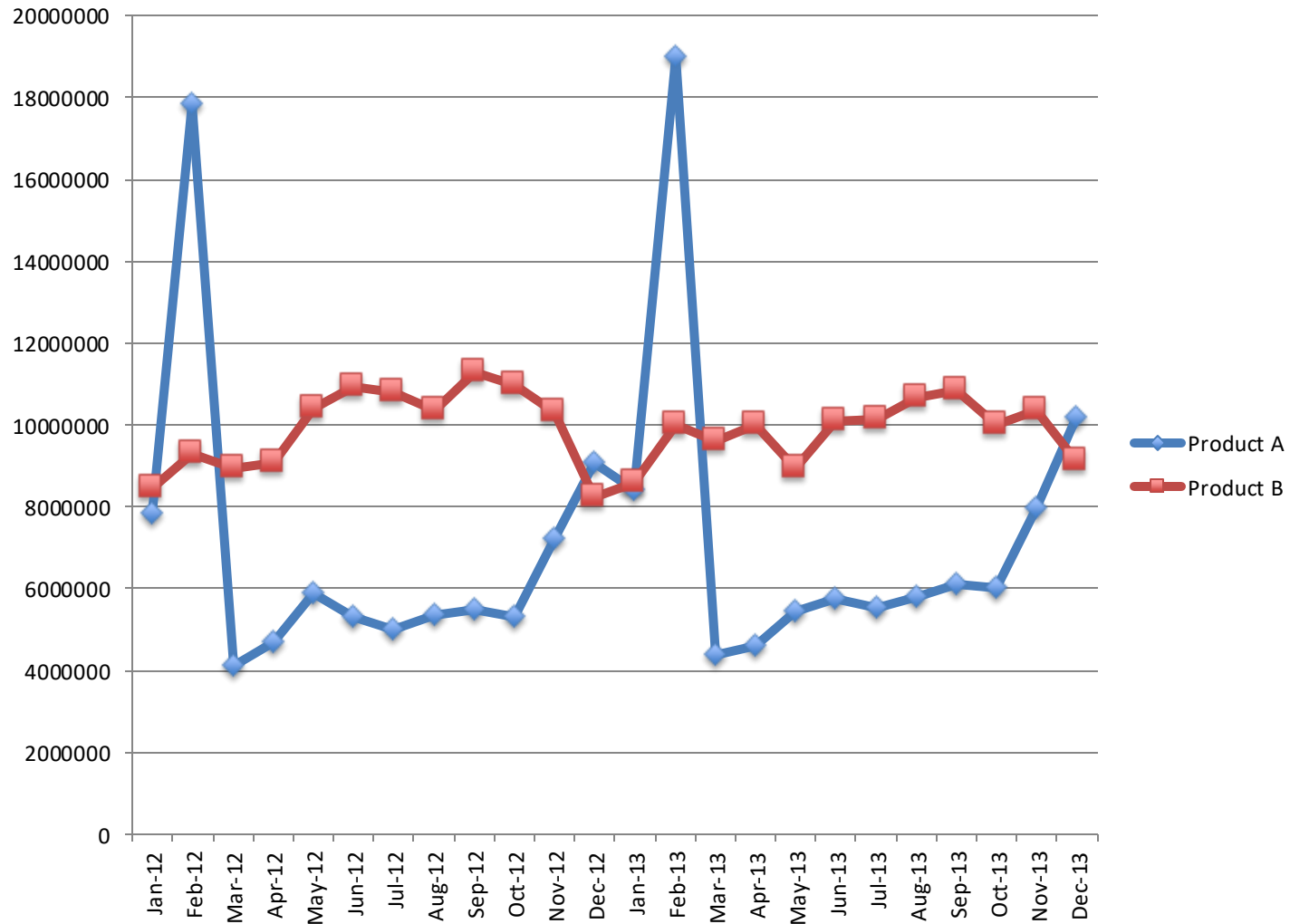
- If you still can't see...



Bar Charts and Time Series Plots

- Ans: Product A is Hershey's Kisses – spikes of sales in February (Valentine's day)
- Small bump in sale around Christmas for Hershey's Kisses – may be given as gifts
- Sometimes, *time series* plots are also drawn

Bar Charts and Time Series Plots



Histograms

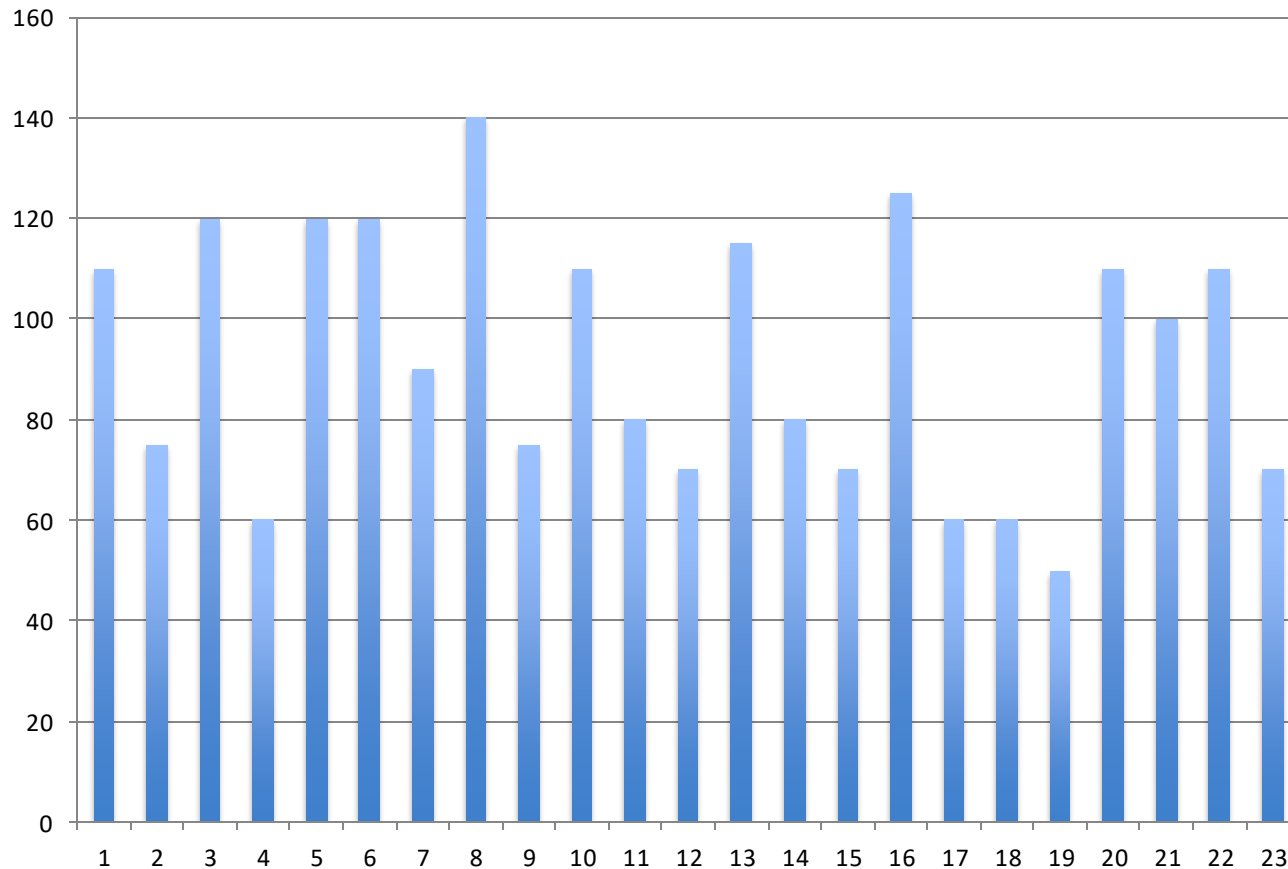
- **Histograms** help you see where data are concentrated and where they are spread out
- Consider the following example. 23 MBA students were confidentially surveyed about the level of annual income they anticipate after graduation

110	90	115	50
75	140	80	110
120	75	70	100
60	110	125	110
120	80	60	70
120	70	60	

- Let's try to visualize...

Histograms

- Here is a bar chart (doesn't really seem to help...)



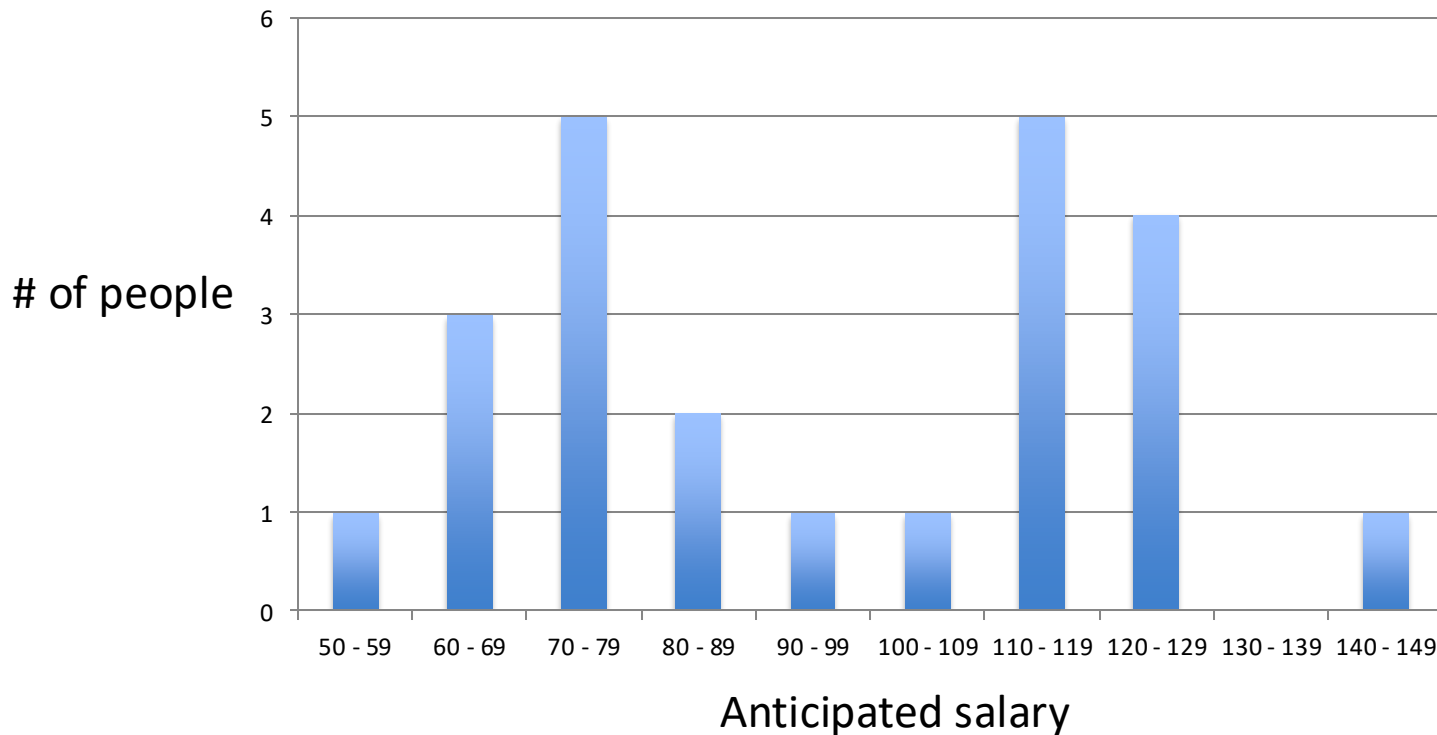
Histograms

- Do a frequency count

				Salary range # of people	
110	90	115	50	50 - 59	1
				60 - 69	3
75	140	80	110	70 - 79	5
				80 - 89	2
120	75	70	100	90 - 99	1
				100 - 109	1
60	110	125	110	110 - 119	5
				120 - 129	4
120	80	60	70	130 - 139	0
				140 - 149	1
120	70	60			

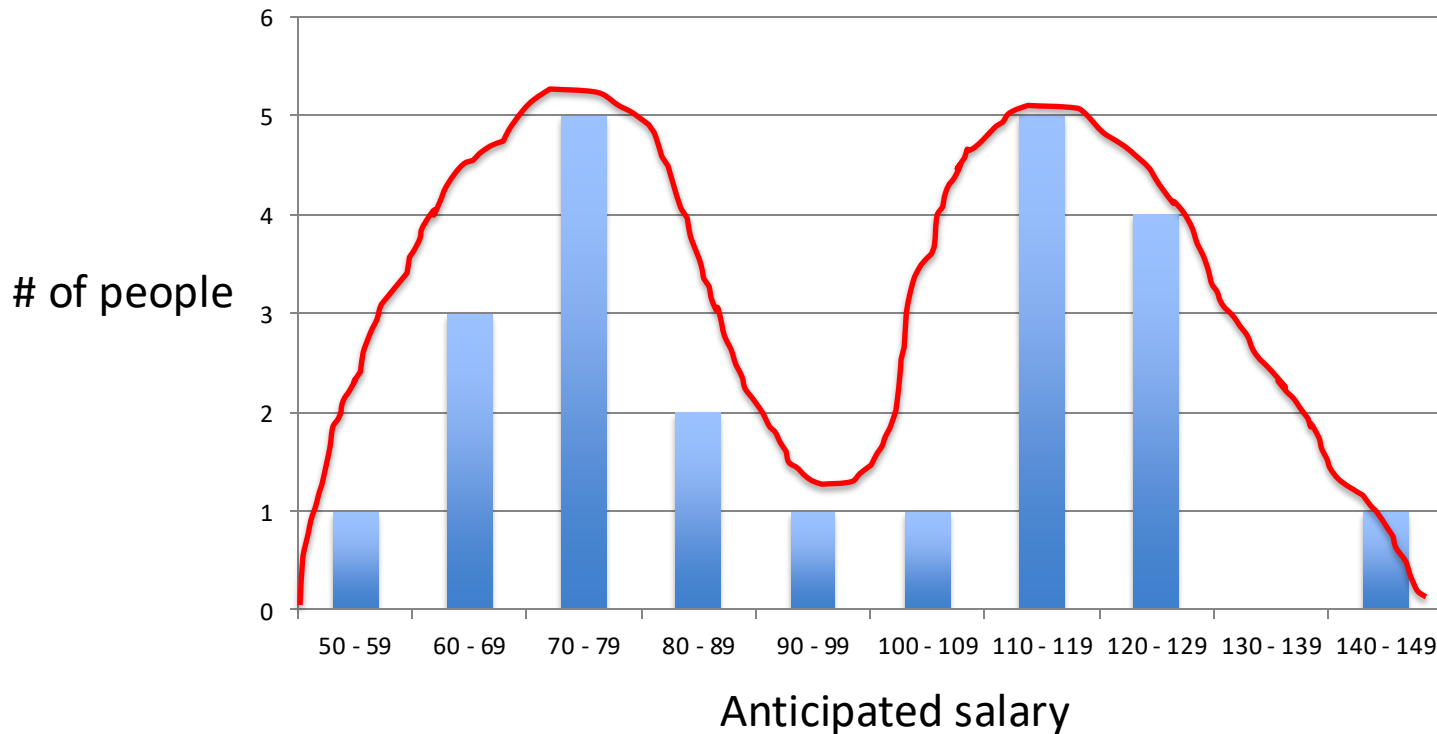
Histograms

- Do a frequency count
- Then make a bar chart of the frequency count
- Now you see two peaks



Histograms

- Do a frequency count
- Then make a bar chart of the frequency count
- Now you see two peaks
- Possibly corresponding to two different industries etc.



Mean, Median and Mode

- Examining data points one by one can be time consuming or impractical $\Rightarrow$ Need to summarize data
- Key metrics: mean, median, mode, standard deviation
- *Mean* is the sum of values divided by the number of them.
- *Median* is the middle number when list of values are arranged from lowest to highest.
- *Mode* is the value that occurs most frequently
- These are measures of the “centrality” of the data

Mean, Median and Mode

- Consider the previous data set.

110	90	115	50
75	140	80	110
120	75	70	100
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appear most frequently

and then take the 12th position.

(note: for even # of data points, take average of the 2 middle #'s)

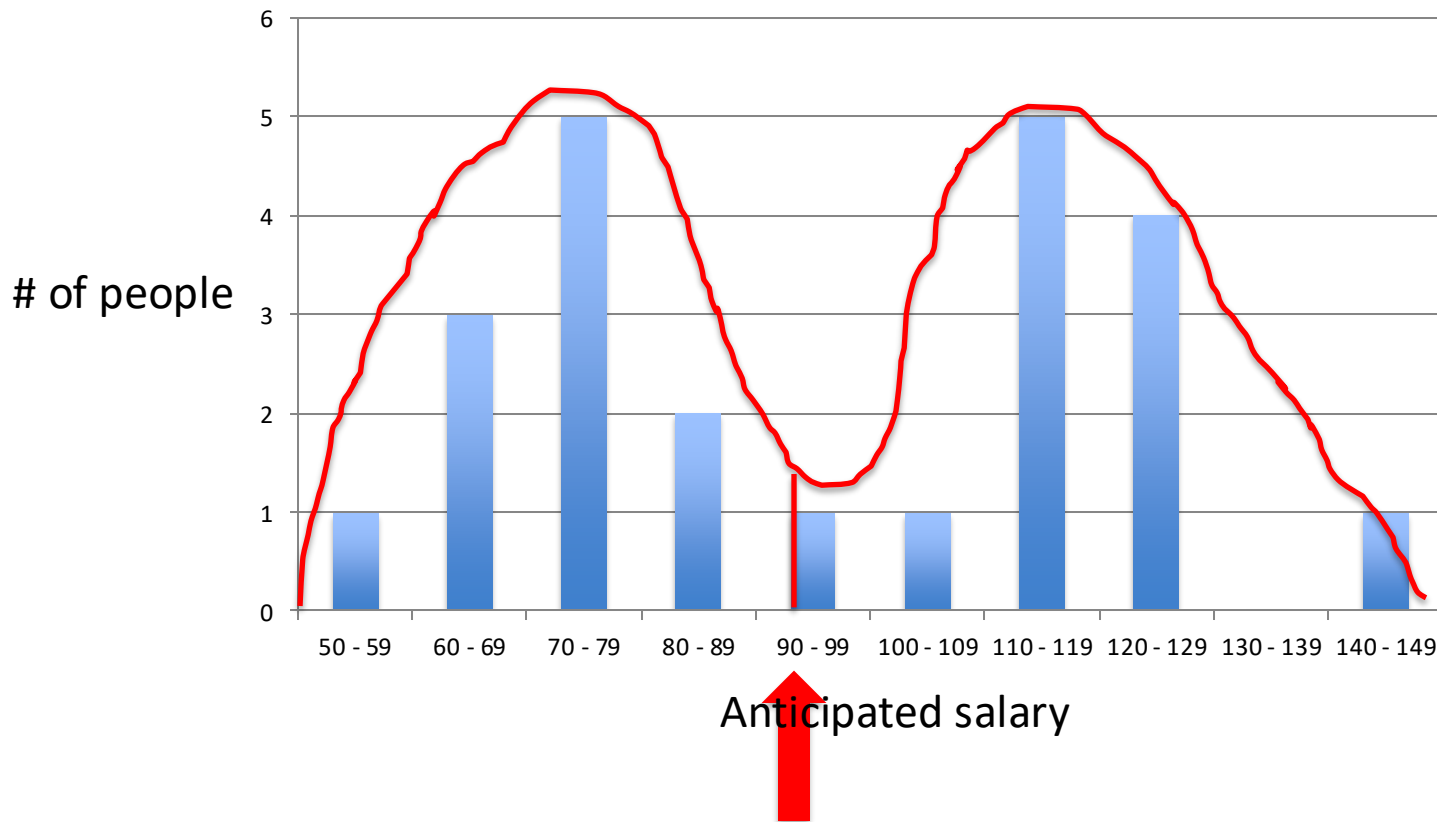
- Mean = 92.17; Median = 90; Mode = 110

$$= \frac{110 + 90 + 115 + 50 + \dots}{23}$$

order from smallest to biggest:
50, 60, 60, 60, 70, ..., 140

Mean, Median and Mode

- Find the median on the histogram.
- The area to the left of median is 50% of the total area
- I.e., median is the 50-percentile point



Mean, Median and Mode

- Which central measure of tendency should be used?

Mean, Median and Mode

- Which central measure of tendency should be used?
- Mean: doesn't capture the bimodal behavior
- Median: doesn't capture the bimodal behavior
- Mode: the best for this problem, if used correctly..

Mean, Median and Mode

Another example: annual income of households in California

- Typical median income is \$40,000 ~ \$50,000
- Mean income can be much higher; e.g., > \$60,000
- Discrepancy because the richest households can be extremely rich

↑
mean is sensitive to outliers
median is not.

Mean, Median and Mode

Another example: annual income of households in California

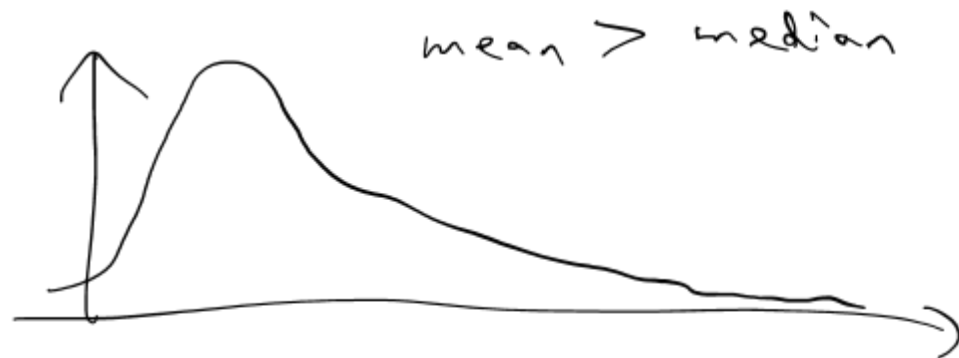
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Generally speaking,

- If the distribution is left-skewed / negatively skewed, median > mean
- If the distribution is right-skewed / positively skewed, median < mean



left-skewed /
negatively skewed



right-skewed /
positively skewed

Standard Deviation

- What about variability of data?
- *Standard deviation* (SD) captures the notion of variability.
- $SD = \sqrt{\text{mean of (deviations from average)}^2}$
- If data values are $X_1, X_2, \dots, X_n$: let

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

be the mean. Then

$$SD = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

measuring the
average distance between
observations and
center of distribution

- Why (deviations from average)²? Because deviations can be both positive or negative.